LESSON 01 / 08

A point is a pair of signed coordinates

A point $P = (x, y)$ is reached by moving x units horizontally and y vertically from the origin. Coordinate differences, not absolute positions, measure displacement.

SHORT DERIVATION

Distance and midpoint from the right triangle

For $A(x_1, y_1)$ and $B(x_2, y_2)$, the horizontal and vertical legs have lengths $|x_2 - x_1|$ and $|y_2 - y_1|$. Pythagoras gives

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

The midpoint has half of each displacement: $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$.

WORKED EXAMPLE

Read displacement before calculating

From $A = (-2, 1)$ to $B = (4, 9)$ the displacement is $(6, 8)$, so $AB = \sqrt{6^2 + 8^2} = 10$. The midpoint is $(1, 5)$.

Section formula: keep the ratio orientation explicit

If P divides AB internally in the ratio $AP : PB = m : n$, then

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n}\right).$$

The coefficient of A is the opposite ratio-part n. This is a weighted average: for $m = n$, it is the midpoint. For external division with $AP : PB = m : n$, use $P = ((mx_2 - nx_1)/(m - n), (my_2 - ny_1)/(m - n))$ with $m \neq n$; verify which endpoint lies between the other two. If $m = n$, no finite point divides externally in that ratio.

WORKED EXAMPLE

A ratio check catches reversed weights

If $A = (1, 2)$, $B = (6, 7)$ and $AP : PB = 2 : 3$, then $P = (3, 4)$: it lies two-fifths of the way from A to B. Substitution verifies $AP : PB = 2 : 3$.

COMMON TRAP

Section-formula trap

Never memorize the numerator without the ratio convention. A point dividing AB in $2 : 3$ must be closer to A, so its coordinates must be weighted more heavily toward A.

Shift the origin, not the geometry

Let the new origin O' have old coordinates (h, k), with axes kept parallel. If a point has new coordinates (X, Y) and old coordinates (x, y), then

$$(x, y) = (X + h, Y + k), \qquad (X, Y) = (x - h, y - k).$$

SHORT DERIVATION

Translation rule for equations

To rewrite an old equation in the new coordinates, substitute $x = X + h$ and $y = Y + k$, then simplify. Example: the old line $2x - y + 1 = 0$ viewed from $O' = (1, -2)$ becomes $2(X + 1) - (Y - 2) + 1 = 0$, i.e. $2X - Y + 5 = 0$. Translation changes constants and point coordinates, not lengths, angles, or slopes.

Centroid is the average of the vertices

For triangle vertices $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$,

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right).$$

The two medians meet at this point; it divides each median in the ratio $2 : 1$ measured from the vertex.

The signed double-area of triangle ABC is

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2).$$

Its absolute value divided by 2 is the area; it vanishes exactly for collinear vertices. This determinant identity is a useful derived coordinate test, not a separate line-syllabus bullet.

KEY RESULT

Coordinate habits

Write displacement as $(x_2 - x_1, y_2 - y_1)$; check an internal section point lies between its endpoints; for a shifted origin substitute old coordinates $x = X + h, y = Y + k$; and average coordinates for the centroid.

LESSON 02 / 08

Slope measures rise per run

For distinct points $A(x_1, y_1), B(x_2, y_2)$ with $x_2 \neq x_1$,

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

A vertical line has no finite slope; a horizontal line has slope 0. Slope is unchanged by translation or by choosing a different pair of points on the same line.

WORKED EXAMPLE

Build one line in several forms

Through $(2, -1)$ with slope 3: $y + 1 = 3(x - 2)$, hence $3x - y - 7 = 0$. Its slope-intercept form is $y = 3x - 7$.

A line has several useful forms

| Form | Equation / use | |---|---| | Point-slope | $y - y_1 = m(x - x_1)$; requires nonvertical line | | Two-point | $(y - y_1)(x_2 - x_1) = (y_2 - y_1)(x - x_1)$; works when points differ | | Slope-intercept | $y = mx + c$; misses vertical lines | | Intercept | $x/a + y/b = 1$; requires both intercepts nonzero | | General | $ax + by + c = 0$, $(a, b) \neq (0, 0)$ | | Normal | $x\cos\alpha + y\sin\alpha = p$; signed perpendicular distance p from the origin |

In $ax + by + c = 0$, (a, b) is a normal vector and a direction vector is $(-b, a)$. If $b \neq 0$, slope is $-a/b$; if $b = 0$, the line is vertical.

SHORT DERIVATION

Normal form from a perpendicular projection

A point $X = (x, y)$ lies on a line at signed distance p from the origin with unit normal $n = (\cos\alpha, \sin\alpha)$ exactly when its projection onto n is p: $X \cdot n = p$. Thus $x\cos\alpha + y\sin\alpha = p$. Keeping p signed avoids changing the line when the normal is reversed.

WORKED EXAMPLE

Vertical lines are not exceptional in general form

The vertical line through $(4, -2)$ is $x = 4$, or $x - 4 = 0$. It has no slope-intercept form, but it is fully represented by $ax + by + c = 0$.

COMMON TRAP

Intercept-form trap

$x/a + y/b = 1$ presumes $a, b \neq 0$. A line through the origin has zero intercepts and should be written in point-slope or general form instead.

LESSON 03 / 08

Compare directions without trusting a sketch

For nonvertical slopes m_1, m_2, the acute angle θ between the lines satisfies

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|.$$

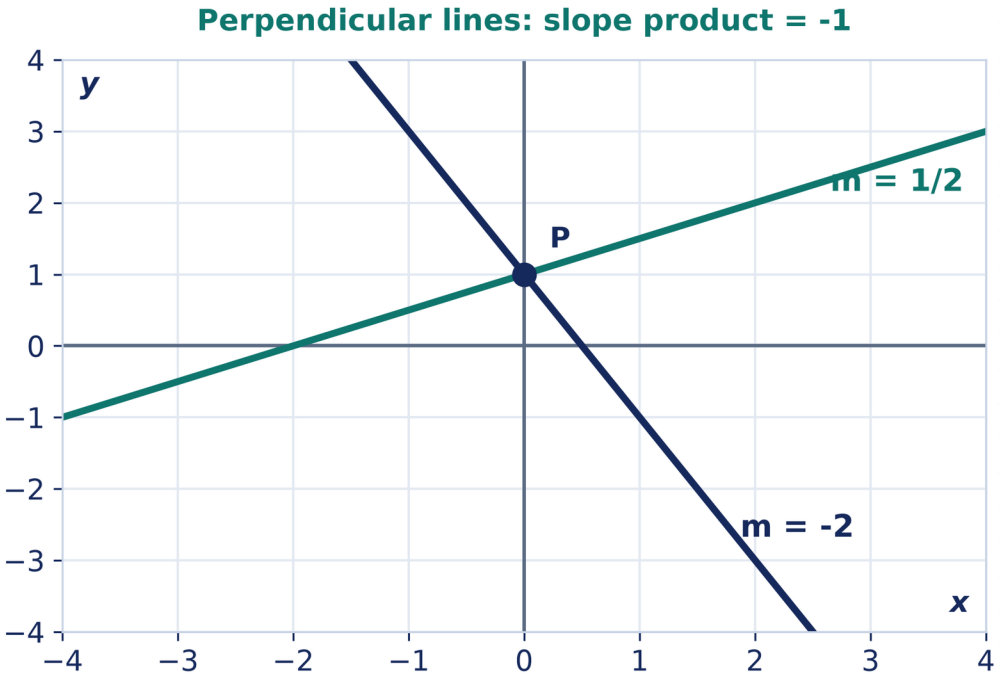
Hence parallel lines have equal slopes and perpendicular lines have $m_1 m_2 = -1$. The slope test needs care for vertical lines; the coefficient test below does not.

For $L_i : a_i x + b_i y + c_i = 0$, use normals $n_i = (a_i, b_i)$. The acute angle obeys $\cos\theta = \frac{|a_1 a_2 + b_1 b_2|}{\sqrt{a_1^2 + b_1^2}\sqrt{a_2^2 + b_2^2}}$, $\sin\theta = \frac{|a_1 b_2 - a_2 b_1|}{\sqrt{a_1^2 + b_1^2}\sqrt{a_2^2 + b_2^2}}$. Parallel iff $a_1 b_2 - a_2 b_1 = 0$; perpendicular iff $a_1 a_2 + b_1 b_2 = 0$.

WORKED EXAMPLE

A vertical–horizontal pair

$x = 2$ has normal $(1, 0)$ and $y = -3$ has normal $(0, 1)$. Their dot product is 0, so the lines are perpendicular even though one slope is undefined.



Perpendicular distance is a projection

For $L : ax + by + c = 0$ and point $P = (x_0, y_0)$,

$$d(P, L) = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}.$$

The signed distance is the same expression without absolute values, once the normal orientation is fixed.

SHORT DERIVATION

Foot of the perpendicular

Let $H = P - t(a, b)$. Requiring H on L gives $t = (ax_0 + by_0 + c)/(a^2 + b^2)$. Thus

$$H = P - \frac{ax_0 + by_0 + c}{a^2 + b^2}(a, b).$$

The removed vector is parallel to the normal, so $PH \perp L$; its length is the point–line distance.

WORKED EXAMPLE

Distance and foot for the same line

From $P = (1, 2)$ to $x + y = 0$, signed distance is $3/\sqrt{2}$ and $H = (1, 2) - (3/2)(1, 1) = (-1/2, 1/2)$. Check H satisfies $x + y = 0$ and PH has direction $(1, 1)$, normal to the line.

COMMON TRAP

Distance formula is scale-invariant

Multiplying a, b, c by a nonzero number does not change the line or its distance: numerator and denominator scale by the same absolute factor. Do not omit $\sqrt{a^2 + b^2}$.

LESSON 04 / 08

Every line through a fixed intersection

Let $L_1 = 0$ and $L_2 = 0$ be distinct, nonparallel lines meeting at P. Every line through P can be written

$$\lambda L_1 + \mu L_2 = 0, \qquad (\lambda, \mu) \neq (0, 0).$$

Using both coefficients is essential: $L_1 + tL_2 = 0$ alone misses $L_2 = 0$ for finite t.

WORKED EXAMPLE

A pencil through a known point

For $L_1 = x + y - 2 = 0$ and $L_2 = x - y = 0$, the intersection is $(1, 1)$. Every line through it is $\lambda(x + y - 2) + \mu(x - y) = 0$. Choosing $(\lambda, \mu) = (1, 0)$ or $(0, 1)$ recovers the two original lines.

Concurrency test for three lines

For three lines $a_i x + b_i y + c_i = 0$, the determinant

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

vanishes exactly when their coefficient rows are linearly dependent. This means the lines have a common finite point **or** are parallel (including coincident degeneracies). To assert finite concurrency, also check that at least two distinct direction vectors are not parallel.

WORKED EXAMPLE

Check, then locate

$x + y - 2 = 0$, $x - y = 0$, $2x - 3 = 0$ meet at $(1, 1)$: the first two give that point, and the third also holds there. Solving two lines and checking the third is the safest concrete test.

Angle bisectors compare signed distances

For two intersecting lines $L_i = a_i x + b_i y + c_i = 0$, their bisectors satisfy

$$\frac{L_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{L_2}{\sqrt{a_2^2 + b_2^2}}.$$

The two signed normalized expressions are signed distances; equal absolute distances define the two bisectors. The $+$ and $-$ choices select the two angle regions.

SHORT DERIVATION

Why normalization is compulsory

The distance from (x, y) to $L_i = 0$ is $|L_i|/\sqrt{a_i^2 + b_i^2}$. Equidistance means these magnitudes agree, which factors into the two signed equations above. Without the denominators, coefficients change the scale and the resulting lines are generally not bisectors.

COMMON TRAP

Three distinct ideas

A line through the intersection of two lines is a pencil; concurrency asks whether several lines share a point; angle bisectors require distance normalization. Do not substitute one test for another.

LESSON 05 / 08

Centroid G: average the vertices

For A, B, C with position pairs (x_i, y_i),

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right).$$

It lies on all three medians and divides each median vertex-to-midpoint in ratio $2 : 1$.

Circumcentre O: equidistant from the vertices

Set $OA^2 = OB^2$ and $OA^2 = OC^2$. The squared terms in the unknown coordinates cancel, leaving two linear equations: the perpendicular bisectors. Their intersection is O; $R = OA$ is the circumradius.

WORKED EXAMPLE

The 3–4–5 right triangle

Take $A = (0, 0), B = (4, 0), C = (0, 3)$. The perpendicular bisector of AB is $x = 2$ and of AC is $y = 3/2$, so $O = (2, 3/2)$ and $R = 5/2$. The hypotenuse midpoint confirms the same centre.

Orthocentre H: intersect the altitudes

An altitude through A is perpendicular to BC, hence has direction normal to BC. Intersect two altitudes to find H. A useful vector identity (coordinates measured from any common origin) is

$$H = A + B + C - 2O.$$

Indeed, $(H - A) = (B - O) + (C - O)$, whose dot product with $C - B$ is $|C - O|^2 - |B - O|^2 = 0$; cyclically the other altitudes follow.

For the right triangle above, $H = A = (0, 0)$: the legs themselves are altitudes.

Incentre I: side-length weighted average

Let $a = |BC|$, $b = |CA|$, $c = |AB|$. The coordinate formula is

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c}\right).$$

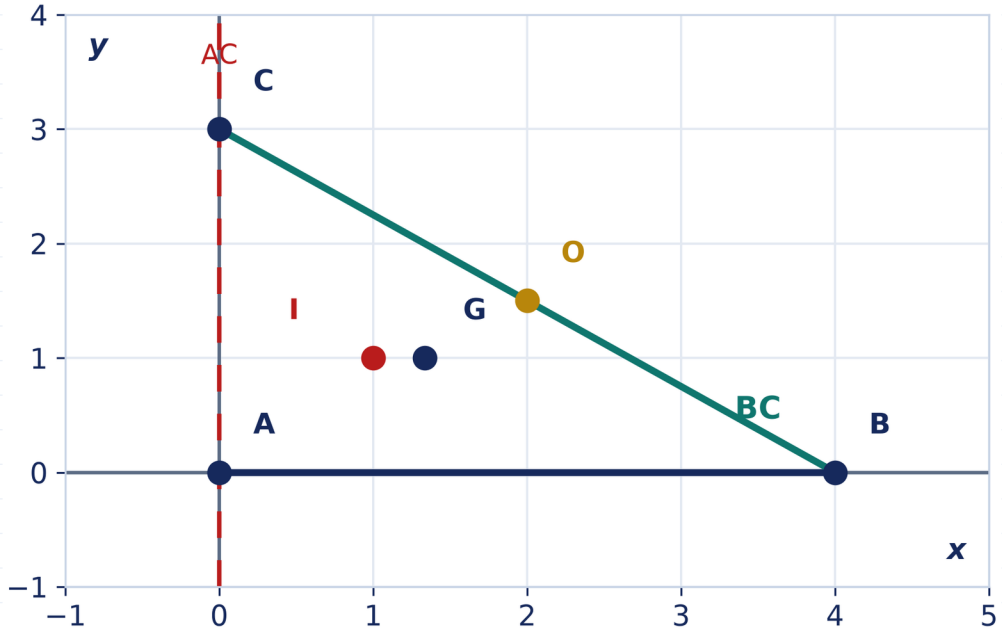
The weights come from the three subtriangle areas: if the common perpendicular distance is r, then $[IBC] = ar/2$, $[ICA] = br/2$, $[IAB] = cr/2$, while $[ABC] = (a + b + c)r/2$. The signed-area coordinates of a point are weighted by these opposite subtriangle areas, giving the formula. Thus its perpendicular distances from the sides are equal; their common value is the inradius.

WORKED EXAMPLE

Centers of the 3–4–5 right triangle

For $A=(0,0), B=(4,0), C=(0,3)$, side lengths opposite A, B, C are $5, 3, 4$. Thus $G=(4/3,1)$, $O=(2,3/2)$, $H=(0,0)$, and $I=(1,1)$. Check I is one unit from each leg; its distance to $3x+4y=12$ is $|3+4-12|/5=1$.

Right triangle: centroid, circumcentre, incentre



KEY RESULT

Use the geometric definition to choose the equations

Centroid: medians or coordinate average. Circumcentre: equal squared distances. Orthocentre: perpendicular altitudes (or the proved vector identity). Incentre: equal perpendicular distances / side-length weights.

LESSON 06 / 08

A locus is a set of points satisfying one condition

Let the moving point be $P=(x,y)$. Write the geometric condition using distances, slopes or projections; simplify to an equation; then check that every point of the resulting equation satisfies the original condition. Squaring can introduce false branches, so the final check matters.

WORKED EXAMPLE

Equidistant from two fixed points

For $A=(0,0)$ and $B=(4,0)$, $PA=PB$ gives $x^2+y^2=(x-4)^2+y^2$, hence $x=2$. This is the perpendicular bisector of AB, not a circle.

Constant distance from one point

$PA=r$ becomes $(x-a)^2+(y-b)^2=r^2$: a circle centred at $A=(a,b)$. The special case $r=0$ is the single point A; $r<0$ has no solutions because distance is nonnegative.

Constant ratio of distances: Apollonius

For distinct fixed points $A=(x_1,y_1)$ and $B=(x_2,y_2)$, let $PA/PB=k\geq 0$. Squaring gives

$$(x-x_1)^2+(y-y_1)^2=k^2\big((x-x_2)^2+(y-y_2)^2\big).$$

If $k=1$, equal quadratic terms cancel and the locus is the perpendicular bisector. If $k\neq 1$, it is a circle with centre

$$\left(\frac{x_1-k^2x_2}{1-k^2},\ \frac{y_1-k^2y_2}{1-k^2}\right),$$

and radius $\dfrac{kAB}{|1-k^2|}$. At $k=0$, interpret the zero-radius circle as the point A.

SHORT DERIVATION

The centre and radius are not guessed

After expansion, the coefficient of x^2+y^2 is $1-k^2$ and the linear terms are $2(k^2x_2-x_1)x+2(k^2y_2-y_1)y$. Divide by $1-k^2$ and complete squares. For $A=(0,0), B=(d,0)$ the centre is $(k^2d/(k^2-1),0)$; substituting it gives radius $kd/|k^2-1|$.

WORKED EXAMPLE

Ratio 2 on a 4-unit baseline

For $A=(0,0)$, $B=(4,0)$ and $PA/PB=2$, $x^2+y^2=4((x-4)^2+y^2)$, so $(x-16/3)^2+y^2=64/9$. Check $P=(16/3,8/3)$: $PA/PB=2$.

COMMON TRAP

Locus checklist

Preserve nonnegative distance conditions when squaring; handle $k = 1$ separately before dividing by $1 - k^2$; distinguish a perpendicular bisector (line) from a ratio circle; and test a point in the final equation against the original condition.

LESSON 07 / 08

A homogeneous quadratic may be two lines

The equation $ax^2 + 2hxy + by^2 = 0$ passes through the origin and is homogeneous. Put $y = mx$ to obtain $bm^2 + 2hm + a = 0$; its real roots are the slopes of the line factors. If $b = 0$, factor in the other direction so a vertical factor is not lost.

SHORT DERIVATION

Angle and perpendicularity from the roots

For two distinct real slopes m_1, m_2, their angle obeys $\tan\theta = |(m_1 - m_2)/(1 + m_1 m_2)|$. The product of roots is a/b when $b \neq 0$, so perpendicular factors require $a/b = -1$, equivalently $a + b = 0$. The discriminant $4(h^2 - ab)$ determines whether the factors are real and distinct, repeated, or complex.

WORKED EXAMPLE

Factor first, then identify the geometry

$2x^2 + 3xy - 2y^2 = (2x - y)(x + 2y)$, so the lines are $y = 2x$ and $y = -x/2$. Their slopes multiply to -1: they are perpendicular.

COMMON TRAP

Scope and degeneracy

This homogeneous-pair criterion is a useful derived tool, not a separately listed line-form bullet in the official syllabus. A real pair requires $h^2 - ab \geq 0$; if equality holds the line is repeated. The slope-root method must account for a vertical factor.

Five checks before a coordinate answer is final

1. **Domain of the formula:** is a denominator zero, or is a slope vertical?
2. **Geometry:** do distances stay nonnegative and are lines distinct where assumed?
3. **Scale:** is a point–line formula invariant if the equation is rescaled?
4. **Locus:** does the equation imply the original condition as well as follow from it?
5. **Triangle centre:** does the proposed point satisfy its defining equal-distance or perpendicularity property?

WORKED EXAMPLE

Quick independent check

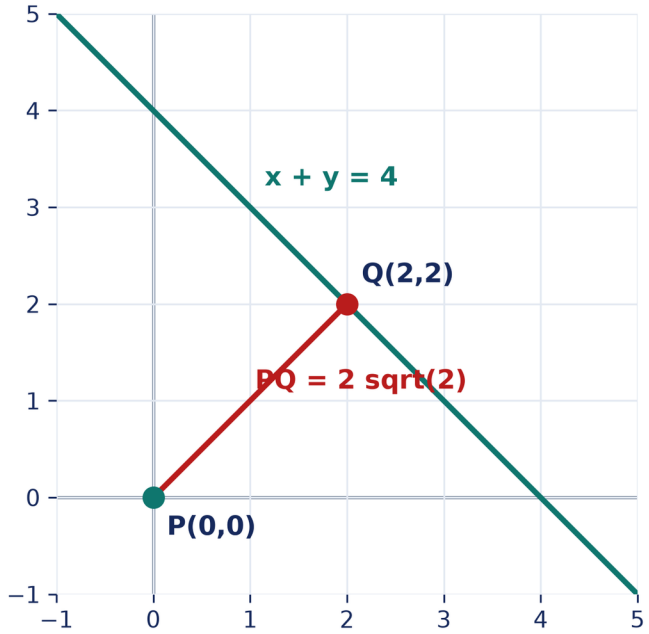
For $A = (0, 0), B = (4, 0)$, the point $P = (2, 3)$ is equidistant from both: $PA^2 = 2^2 + 3^2 = 13 = PB^2$. It lies on $x = 2$, the derived locus. The converse also holds for every point on that line.

LESSON 08 / 08

Distance from a point to a line is perpendicular distance

For $P=(0,0)$ and $x+y=4$, the perpendicular foot is $Q=(2,2)$. Hence $PQ=\sqrt{2^2+2^2}=2\sqrt{2}$, matching $|0+0-4|/\sqrt{1^2+1^2}=2\sqrt{2}$.

The shortest segment from P to the line is perpendicular to it

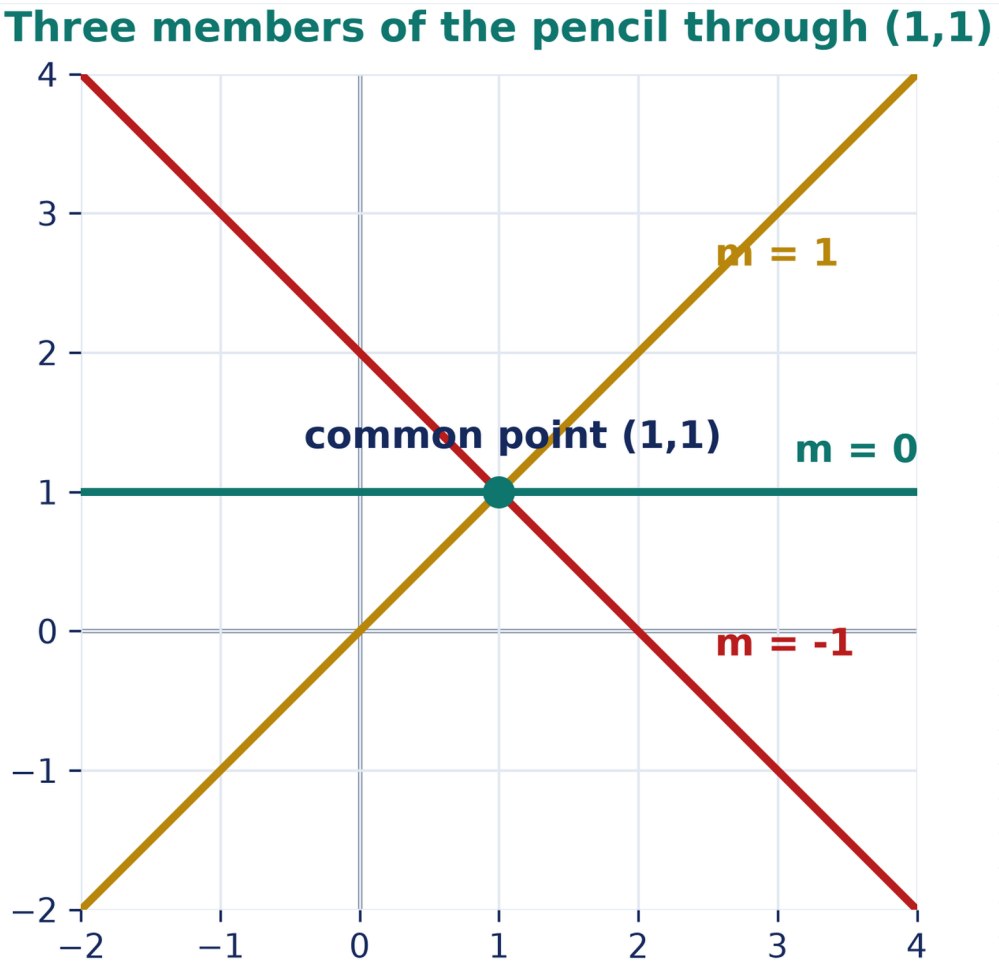


READ THE FIGURE
The line has slope -1 and PQ has slope 1, so their product is -1. Any other segment from P to the line is longer than the perpendicular segment.

The formula $|ax_0+by_0+c|/\sqrt{a^2+b^2}$ includes an absolute value because distance is nonnegative. The sign before taking the modulus instead tells which side of the line the point lies on.

A pencil of lines shares one point

The equation $y-1=m(x-1)$ describes a nonvertical line through $(1,1)$ for every slope m. Different values of m rotate the line around the same point. The vertical line $x=1$ must be added separately if the full pencil is required.



READ THE FIGURE

A linear combination of two line equations L1 + lambda L2 = 0 passes through every common solution of L1 = 0 and L2 = 0. This is the algebraic form of a pencil.

Concurrency questions often become shorter when a pencil is used: impose one additional point or tangency condition to determine the parameter, rather than solving two line equations repeatedly.

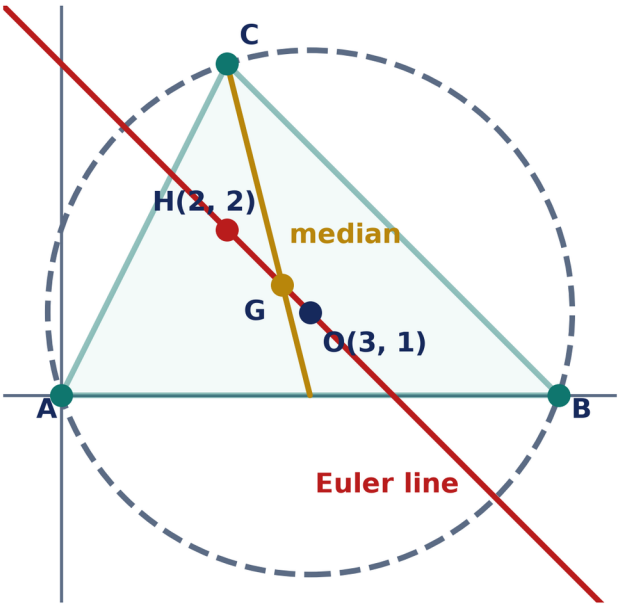
Triangle centres are coordinate averages or intersections

For $A=(0,0), B=(6,0), C=(2,4)$, the centroid is

$$G=\frac{A+B+C}{3}=\left(\frac{8}{3},\frac{4}{3}\right).$$

The perpendicular bisector of AB is $x=3$. Intersecting it with the perpendicular bisector of AC gives circumcentre $O=(3,1)$. The altitudes give orthocentre $H=(2,2)$.

Circumcentre, centroid and orthocentre lie on the Euler line



READ THE FIGURE
The circumcircle confirms O. On the Euler line, GH = 2 OG, with G between O and H. The median from C passes through G.

The centroid always uses equal vertex weights. The incentre instead uses side-length weights, $I=(aA+bB+cC)/(a+b+c)$; do not confuse these two weighted averages.

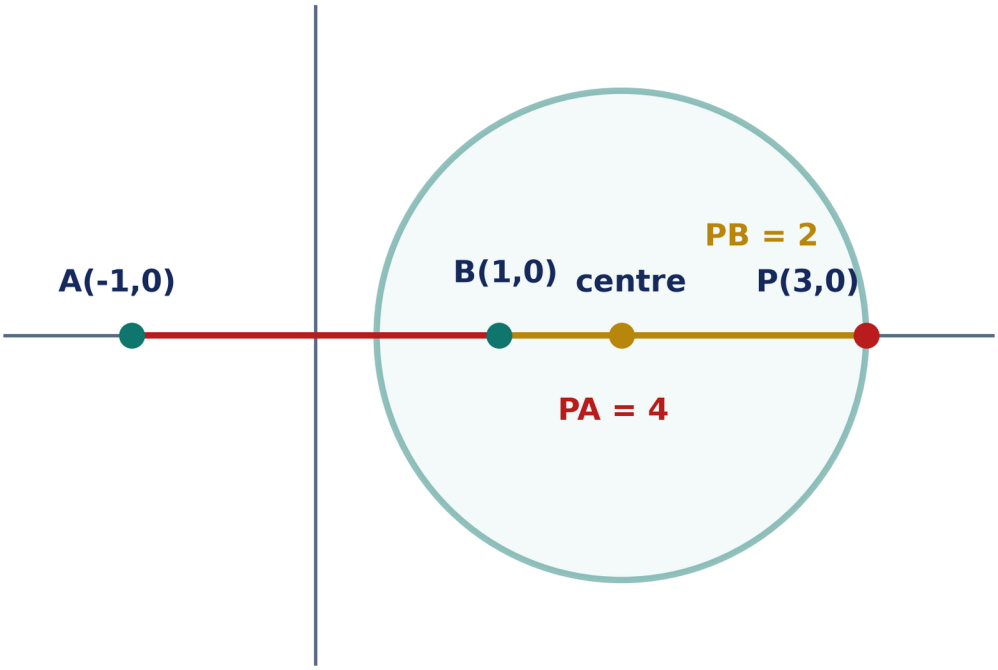
A locus must include construction and verification

Find the locus of $P=(x,y)$ satisfying $PA=2PB$, where $A=(-1,0)$ and $B=(1,0)$.
Squaring the distances gives

$$(x+1)^2+y^2=4\big((x-1)^2+y^2\big).$$

After simplification and completion of squares, $(x-5/3)^2+y^2=(4/3)^2$.

The distance-ratio condition PA = 2 PB gives a circle

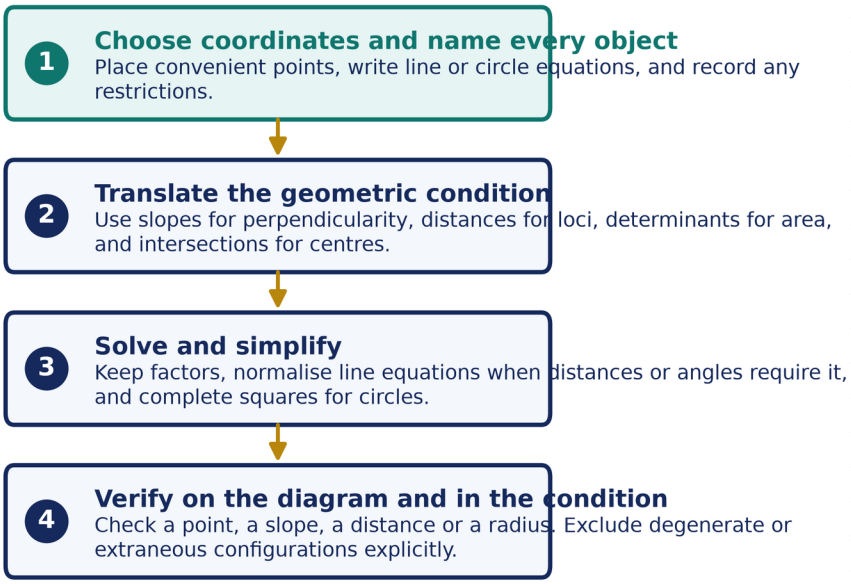


READ THE FIGURE
At P = (3,0), PA = 4 and PB = 2, so the defining ratio holds. The other x-axis intersection, P = (1/3,0), also satisfies it.

Squaring was reversible because both sides were nonnegative. In a general locus problem, check whether an operation introduced or removed points, and test simple points on the final curve against the original condition.

Coordinate-geometry workflow

Translate geometry into algebra without losing the geometry



READ THE FIGURE

Coordinates are a method for expressing geometry, not a replacement for geometric reasoning. A quick diagram often reveals a wrong sign or an impossible centre immediately.

KEY RESULT

Normalise before comparing

For angle bisectors and distances, scale matters. If $ax + by + c = 0$ is multiplied by a nonzero constant, its geometric line is unchanged but an unnormalised signed expression changes. Divide by $\sqrt{a^2 + b^2}$ before comparing distances.